SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2024-2025

Program: B Tech/MBA Tech/ B Tech Intg Stream ; (CSE-DS/EXTC/MXTC/COMP/CS/DS)

Year:II/III Semester: III/V

Subject: Management Accounting for Engineers

Time: 3 hrs (10 am to 1pm)

Date:10 / 03 / 2025 .

No. of Pages:4

Marks: 100

Re-Examination (Batch 2024-25)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-2 ; SO- ; BL-Understand</td><td>a.</td><td>Give one example of the following:
<table>
<tr><td>i. Variable Cost</td><td>ii. Sunk Cost</td></tr>
<tr><td>iii. Fixed Cost</td><td>iv. Selling and Distribution Expenses</td></tr>
<tr><td>v. Semi-variable Cost</td><td>vi. Administrative Cost</td></tr>
<tr><td>vii. Direct Expenses</td><td>viii. Controllable Cost</td></tr>
<tr><td>ix. Indirect Expenses</td><td>x. Uncontrollable Cost</td></tr>
</table></td><td>5</td></tr>
<tr><td>CO- 3; SO- ; BL-Apply</td><td>b.</td><td>You are furnished with following information:
<table>
<tr><td>Particulars</td><td>Amount</td></tr>
<tr><td>Fixed cost</td><td>100000</td></tr>
<tr><td>Variable cost per unit</td><td>Rs. 100</td></tr>
<tr><td>Selling price per unit</td><td>Rs. 200</td></tr>
<tr><td>Actual units sold</td><td>1500 units</td></tr>
</table>
Determine
1. BEP in units and Rs.</td><td>5</td></tr>
</table>

<table>
<tr><td></td><td></td><td>2. Margin of safety in units and Rs.</td><td></td></tr>
<tr><td>CO- 4; SO-; BL-Apply</td><td>c.</td><td>A company established the following direct materials cost standards for one unit of product K.
Standard quantity - 4 Kg; Standard rate – Rs. 5 per Kg; Standard cost – Rs. 20
During the month of January, the company purchased 25,000 kg of materials at a cost of Rs. 128,750 and used 10,250 kg of materials to produce 2500 units of product K.
A) Calculate direct materials price variance
B) Calculate direct materials usage variance</td><td>5</td></tr>
<tr><td>CO-1 ; SO-; BL-Understand</td><td>d.</td><td>How Cost Accounting is different from Financial Accounting?</td><td>5</td></tr>
<tr><td>Q2
CO-3; SO-; BL-Analyze</td><td>a</td><td>• M/s Tubes Ltd. are the manufacturers of picture tubes for T.V. The following are the details of their operation during 2017:
• Annual market demand –5200 tubes
• Ordering cost = Rs. 100 per order
• Inventory carrying cost = 20% per annum
• Cost of tubes = Rs. 500 per tube

Compute from the above:
economic order quantity. If the supplier is willing to supply quarterly tubes - 1500 units at a discount of 5%, is it worth accepting?</td><td>15</td></tr>
<tr><td>Q2
CO-1; SO-; BL-Understand</td><td>b</td><td>Explain the concept of Opportunity cost and how is it different from accounting cost. Give 2 examples explaining opportunity cost.</td><td>5</td></tr>
<tr><td>Q3
CO-3; SO-; BL-Analyze</td><td></td><td>The sales turnover and profit during two years were as follows:
<table><tr><th>Year</th><th>Sales (₹)</th><th>Profit (₹)</th></tr><tr><td>2020</td><td>3,00,000</td><td>40,000</td></tr><tr><td>2021</td><td>3,40,000</td><td>50,000</td></tr></table>You are required to calculate:
i. P/V Ratio
ii. Fixed Costs
iii. Break Even Sales Volume
iv. Calculate Desired Sales required to earn a desired profit of ₹ 1,00,000
V. Calculate profit if Desired Sales are ₹ 6,00,000</td><td>[20]</td></tr>
<tr><td>Q4
CO-2; SO-;</td><td></td><td>Mr. A furnishes the following data related to the manufacture of a standard product during the month of August 2024:
Opening stock of Raw material – Rs.50,000</td><td>[20]</td></tr>
</table>

BL-Apply		Closing stock of Raw material – Rs. 60,000 Opening stock of WIP – Rs. 20,000 Closing stock of WIP- Rs. 25,000 Raw material purchased – Rs. 1,00,000 Import duty on Raw material – Rs. 15,000 Direct labour - Rs 5,000 Direct chargeable expenses – Rs. 10,000 Machine hours worked – 20000 hours Machine hour rate - Rs 2 per hour Foreman's wages – 50,000 Other Administration overheads = 20% of works cost Selling overheads - Rs 0.25 per unit Unit produced - Rs 20,000 Closing stock to be valued at Rs. 11.8 per unit. Unit sold - 16,000 @ Rs 14 per unit You are required to prepare a cost sheet from the above showing - The Total cost, Cost per unit sold and profit for the period and Profit per unit.	
Q5 CO-3; SO-; BL- Analyze	a	Following figures are extracted from the books of a manufacturing concern for the year 2024-25 Direct Materials ₹ 2,05,000; Direct Labour ₹ 75,000; Fixed Overheads ₹ 60,000; Variable Overheads ₹ 1,00,000 ; Sales ₹ 5,00,000 Calculate P/V ratio, Profit, Break Even Point (In Amount) And Margin of Safety (In amount) . What will the effect on Break Even Point of an increase of 10% in – A) Fixed Expenses B) All Variable Expenses without any change in Fixed expenses and sales Solve this a and b part separately	[12]
Q5 CO-3; SO-; BL- Understan d	b	Describe the key steps involved in the budgeting process of a company. Why is each step important for effective budget management?	[8]
Q6 CO-2; SO-; BL- Understan d	a.	Discuss any 2 types of inventory control system along with its two advantages	[8]
Q6	b.	Reli Ltd. gives you the following cost details for manufacture of product Relifast at capacity level of 20,000 units.	[12]

CO-5; SO-; BL-Create

Particulars	₹	Nature of Variability
Direct Material	15,00,000	100% Variable
Direct Labour	10,00,000	100% Variable
Factory Overheads :		
- Lighting and Heating	5,00,000	75% Variable
- Repairs and Maintenance	4,00,000	50% Variable
- Depreciation	6,00,000	100% Fixed
Office Overheads	2,50,000	10% Variable
Selling and Distribution Overheads	4,00,000	50% Variable

Prepare a production cost budget for production of 17,500 units and 30,000 units, showing total cost and unit cost at each level.

Q7

CO-2; SO-; BL-Apply

a. Shrilata Enterprises manufactures a special productFlyer. The following particulars were collected for the year 2024:

(a) Yearly demand of Flyer 24,000 units

(b) Normal usage 250 units per week.

(c) Minimum usage 225 units per week.

(d) Maximum usage 275 units per week.

(e) Re-order period 8 to 10 weeks.

(f) Re-order quantity 386 units

Determine: Re-order level, Minimum level, Maximum level and Average stock level

[8]

Q7

CO-4; SO-; BL-Apply

b. A company established the following direct labour cost standards for one unit of product Y and Product Z.

	Product Y	Product Z
Standard Hours	2 hours	1.5 Hours
Standard Rate	Rs. 25 per hour	Rs. 20 per hour

During the month of July, the company manufactured 10000 units of Product Y in July for which 25,000 direct labour hours were employed. The direct wages related to Product Y in July were Rs. Rs. 575,000. Similarly for Z, 20,000 direct labour hours were worked and 12500 units of product Z were manufactured. The total wages related to direct labour in July were Rs. 405,000.
Calculate the direct labour rate variance, direct labour efficiency variance, the direct labour cost variance for both products.

[12]